

Software Project Proposal

Project Title: Eco-Matic Vending Machine Simulator (WinForms Edition)

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Version: 1.0

1. Introduction

1.1 Project Overview

The "Eco-Matic" Simulator is a WinForms application built with C#. It promotes SDG 12 (Responsible Consumption) by integrating a standard vending machine with a unique "trash-to-credit" recycling system, moving beyond the previous console version to a professional GUI with persistent database connectivity.

1.2 Objectives

- Develop an intuitive WinForms GUI for customer and admin interactions.
- Implement MySQL database persistence to replace csv.
- Simulate a "Recycle for Credit" module for sustainable waste management.
- Enforce role-based security via an authenticated administrator dashboard.

1.3 Scope

The system includes a **Customer UI** for purchasing, recycling, and balance management, and an **Administrator UI** for inventory CRUD (Create, Read, Update, Delete) and transaction auditing.

2. System Analysis

2.1 Problem Statement

The previous CLI version lacked user-friendly interaction and used prone-to-error CSV storage, making data management and complex reporting difficult.

2.2 Proposed Solution

A WinForms C#.NET application connected to MySQL will provide visual feedback, ensure data integrity, and allow for efficient inventory and log management.

2.3 Feasibility Study

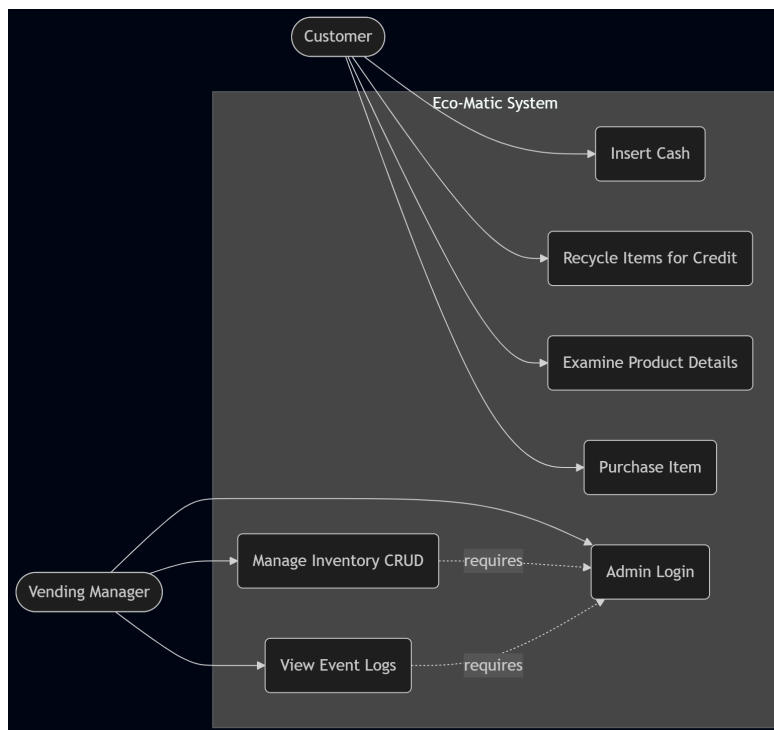
- **Technical:** Reuses core OOP logic from OOP1 with new ADO.NET database handling.
- **Operational:** GUI significantly improves usability over command-line inputs.
- **Financial:** Uses free tools (Visual Studio Community, MySQL).

3. System Design

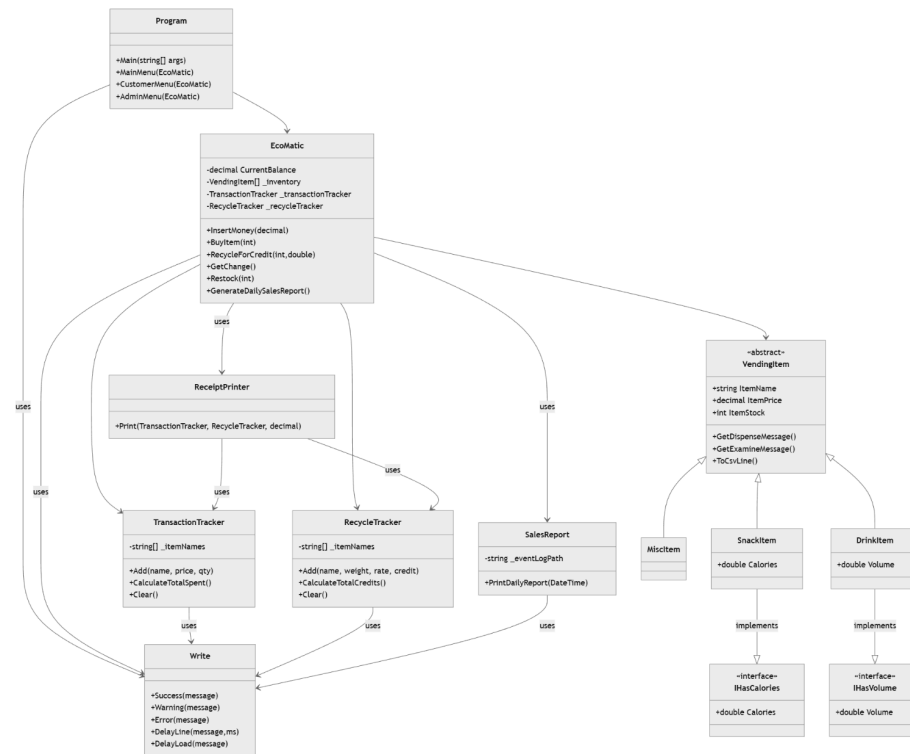
3.1 System Architecture

A Presentation, Business Logic, and Data Access, ensures a clean separation between the WinForms UI and the MySQL database.

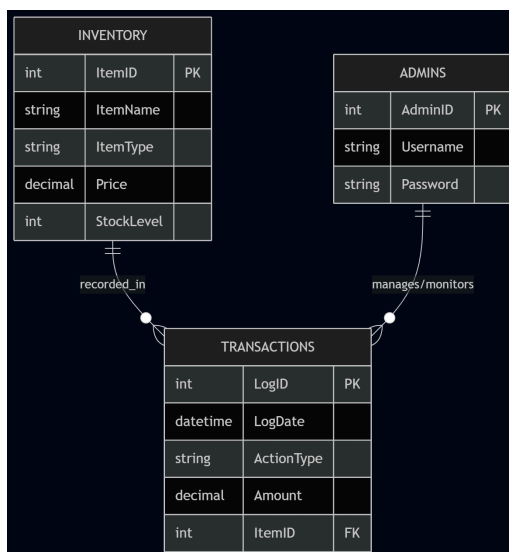
3.2 Use Case Diagram



3.3 Class Diagram



3.4 ER Diagram



4. Implementation Plan

4.1 Development Tools & Technologies

- **C#, WinForms, .NET Core**
- **MySQL**
- **Visual Studio 2022**

4.2 Modules

- **Auth:** Secure Admin login.
- **CRUD:** Inventory management.
- **Transactions:** Customer balance and recycling logic.

5. Database Design

5.1 Tables

Inventory	
Column Name	Data Type
ItemID	INT
ItemName	VARCHAR(50)
ItemType	VARCHAR(20)
Price	DECIMAL(10,2)
StockLevel	INT

Transaction	
Column Name	Data Type
LogID	INT
LogDate	DATETIME
ActionType	VARCHAR(30)
ItemID	INT

Admin	
Column Name	Data Type
AdminID	INT
Username	VARCHAR(50)
Password	VARCHAR(50)

5.2 Sample Queries

- **Restock:** `UPDATE Inventory SET StockLevel = StockLevel + 10 WHERE ItemID = 1;`
- **Log:** `INSERT INTO Transactions (ActionType, ItemID) VALUES ('Purchase', 1);`

6. Testing & Quality Assurance

- **Unit/Integration:** Testing change logic and database connectivity.
- **User Acceptance Testing:** Verifying intuitive UI flow and error handling.
- **Outcome:** A stable, secure desktop app with zero data loss.

7. Conclusion

Eco-Matic WinForms Edition improves usability and reliability by transitioning to a GUI and MySQL database, showcasing advanced OOP and professional data management.

8. References

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